

Practice Problem Set -06 (Matrix-II)

Q1. Verify Cayley – Hamilton theorem and express $A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2$ as a linear polynomial in A, if $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$.

Ans: $-4A + 5I$

Q2. Verify Cayley – Hamilton theorem and hence find A^{-1} , where $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$.

Ans: $A^{-1} = \frac{1}{3} \begin{bmatrix} -1 & 2 & 0 \\ 2 & -5 & 2 \\ 0 & 2 & -1 \end{bmatrix}$

Q3. Find the latent root and latent vectors and modal matrix B for the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$, Also find the spectral radius of matrix A.

Ans: latent root are $-3, -3, 5$

$$B = \begin{bmatrix} 3 & 2 & 1 \\ 0 & -1 & 2 \\ 1 & 0 & -1 \end{bmatrix}$$

spectral radius = 5

Q4. Verify Cayley – Hamilton theorem for matrix

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \text{ and hence find } A^{-1}.$$

Ans: $A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$

Q5. Find eigen values, eigen vectors and modal matrix B for matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.

Ans: $B = \begin{bmatrix} 1 & 1 & 3 \\ -1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

Q6. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$, find the eigen values of $3A^3 + 5A^2 - 6A - 2I$.

Ans: 0, 106, 6



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Q7. If $A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$, find the eigen values of $A^3 + 5A^2 - 2A -$

5I. Also find the spectral radius of A.

Ans: Eigen values are 19, 1, -1 and spectral radius = 2